Bayesian Inference

TM Chapter 6

Outline

- Bayes Theorem
- MAP & ML hypotheses
- MAP learners
- Minimum description length (MDL) principle
- Bayes-optimal, Gibbs, and Naive Bayes classifiers
- …

Why Study Bayesian Inference?

Robust to overfitting

Importance:
Confidence level estimation

Provides practical learning algorithms:

- Naive Bayes classifiers & Bayesian belief networks
- Allows prior knowledge (in the form of prior belief) to be combined with observed data to give probabilistic prediction
- Allows new input instance to be classified by combining predictions of multiple hypotheses weighted by their beliefs

as data
streams in ->

- Incrementally updates belief of hypothesis with each training example

You don't kill off any hypothesis

Provides useful conceptual framework:

- Provides "gold standard" to evaluate other learning algorithms
- Additional insight into Occam's razor

incremental form of Bayes' Rule

D_1 D_2 D_3

Approximate inference

use old posterior as your new prior

Bayes' Theorem/Belief Update

Two leading techniques:
Monte Carlo sampling
Stochastic sampling

What's the likelihood that this hypothesis generates the data?

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)} \quad \text{where}$$

= B * P(D/h) * P(h) where P = 1/P(D)

Variational inference
(Uses upper/lower bound)

- $P(h)$: prior belief of hypothesis h

Law of total probability

- $P(D|h)$: likelihood of data D given h

Latent variable?

- $P(D) = \sum_{h \in H} P(D|h)P(h)$: marginal likelihood/evidence of D

expensive to compute/ computationally intractable

3) not enough data to substantiate prior

Trust -> something you can't measure with a sensor

- $P(h|D)$: posterior belief of h given D

Limitations:
1) wrong prior set, take lots of data to erase
2) need to choose likelihood function correctly

Limitations.

Need to specify all distribution of hypothesis

- Requires specifying probabilities and underlying distributions
- Often prohibitively expensive to compute evidence

How to Choose Hypothesis?

We generally want the most probable hypothesis given the training data, i.e., maximum *a posteriori* hypothesis:

$$
\begin{aligned}
h_{\mathrm{MAP}} &= \arg\max_{h \in H} P(h|D) \\
&= \arg\max_{h \in H} \frac{P(D|h)P(h)}{P(D)} \\
&= \arg\max_{h \in H} P(D|h)P(h) \ .
\end{aligned}
$$

Pick LARGEST POSTERIOR BELIEF among all hypothesis

doesn't vary with H

^ un-normalised posterior

If $P(h) = P(h')$ for any $h, h' \in H$, then we can further simplify and choose the maximum likelihood hypothesis:

$$h_{\mathrm{ML}} = \arg\max_{h \in H} P(D|h) \ .$$

From 0.008 -> 0.2 something

Example: Medical Diagnosis

What hypothesis gives you largest posterior belief?

Does the patient have cancer or not?

- A patient takes a lab test and the result comes back +ve.
- The test returns a correct +ve result in only 98% of the cases in which cancer is actually present,
- and a correct –ve result in only 97% of the cases in which cancer is not present.
- Furthermore, 0.008 of the entire population have this cancer.

P(cancer|+) = P(+|cancer)P(cancer) = 0.00784

P(~cancer|+) = P(+| ~cancer) P(~ cancer) = 0.02976

$P(cancer) =$ 0.008

$P(+ \mid cancer) =$ 0.98

$P(+ \mid \neg cancer) =$ 0.03

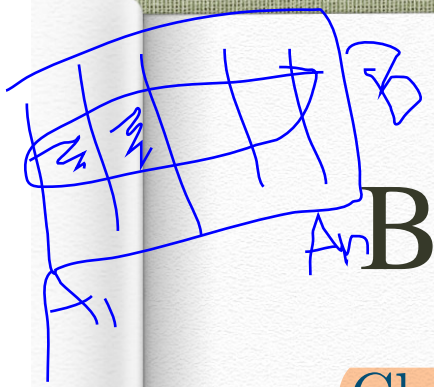
$P(\neg cancer) =$ 0.992

$P(- \mid cancer) =$ 0.02

$P(- \mid \neg cancer) =$ 0.97

Thus, patient ~ cancer

Chance of cancer = 0.02976/ (0.02976+0.00784) = 0.2



B ^ U Ai = U (B ^ Ai)

Basic Probability Formulas

Chain rule for probability. Joint probability $P(A_1, \ldots, A_n)$ of a conjunction of n events $A_1, \ldots, A_n$:

P(A | A1) * P(A1 | A2)

$$P(A_1, \ldots, A_n) = \prod_{i=1}^{n} P(A_i | A_1, \ldots, A_{i-1}) \ .$$

Inclusion-exclusion principle. Probability of a disjunction/ union of n events $A_1, \ldots, A_n$:

P(A or B)
=P(A) + P(B)
-P(A and B)

Conditional independence

$$P(\textstyle\bigcup_{i=1}^{n} A_i) = \sum_{1 \le i \le n} P(A_i) - \sum_{1 \le i < j \le n} P(A_i, A_j) + \sum_{1 \le i < j < k \le n} P(A_i, A_j, A_k) - \ldots + (-1)^{n-1} P(A_1, \ldots, A_n) \ .$$

For efficiency

Marginalization. If events $A_1, \ldots, A_n$ are mutually exclusive s.t. $\sum_{i=1}^{n} P(A_i) = 1$, then $P(B) = \sum_{i=1}^{n} P(B|A_i) P(A_i)$.

Brute-Force MAP Hypothesis Learner

1. For each hypothesis $h \in H$, compute posterior belief

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)} \ .$$

2. Output hypothesis h_{MAP} with highest posterior belief

$$h_{\mathrm{MAP}} = \arg\max_{h \in H} P(h|D) \ .$$

Relation to Concept Learning

Consider our usual concept learning task:

- Input instance space X, hypothesis space H, unknown target concept/function $c : X \rightarrow \{0, 1\} \in H$, noise-free training examples $D = \{d\}$ where $d = \langle \mathbf{x}_d, c(\mathbf{x}_d) \rangle$
- FIND-S outputs most specific hypothesis from version space $VS_{H,D}$

What would Bayes rule produce as the MAP hypothesis?

Does FIND-S output a MAP hypothesis?

Relation to Concept Learning

- Assume that input instances $\mathbf{x}_d$ for $d \in D$ are fixed
- Choose $P(D \mid h)$:

$$P(D|h) = \begin{cases} 1 & \text{if } h \text{ is consistent with } D \\ 0 & \text{otherwise.} \end{cases}$$

Deterministic Likelihood {1,0}

- Choose $P(h)$ to be uniform distribution:

$$P(h) = \frac{1}{|H|} \text{ for all } h \in H.$$

c(xd) = h(x) for all (xd,x) in D

Then,

$$P(h|D) = \begin{cases} \dfrac{1}{|VS_{H,D}|} & \text{if } h \text{ is consistent with } D, \\ 0 & \text{otherwise.} \end{cases}$$

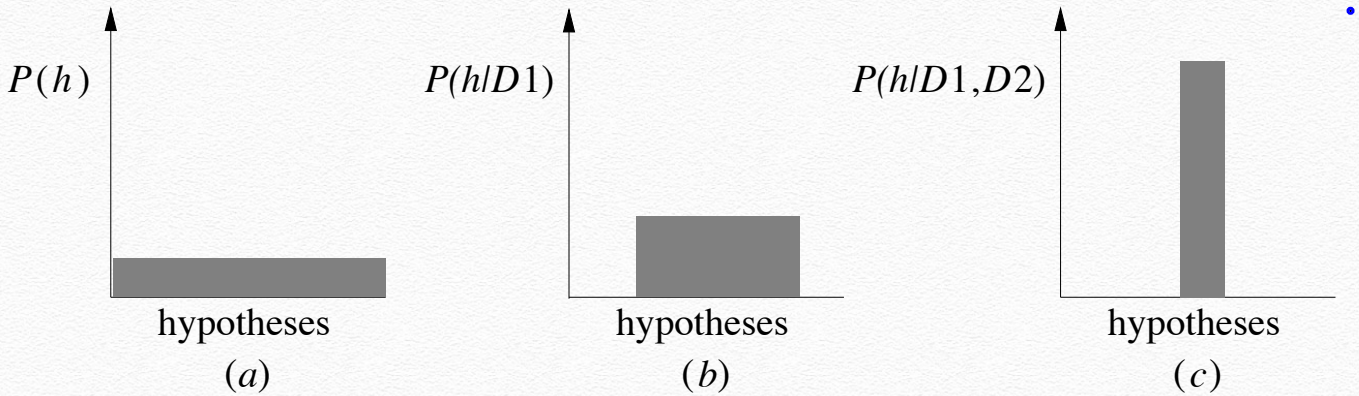
sum to 1

Every consistent hypothesis is a MAP hypothesis!

Belief Update

MAP Hypothesis:
Look at every hypothesis
Compute posterior probability

Limitation: h is too large

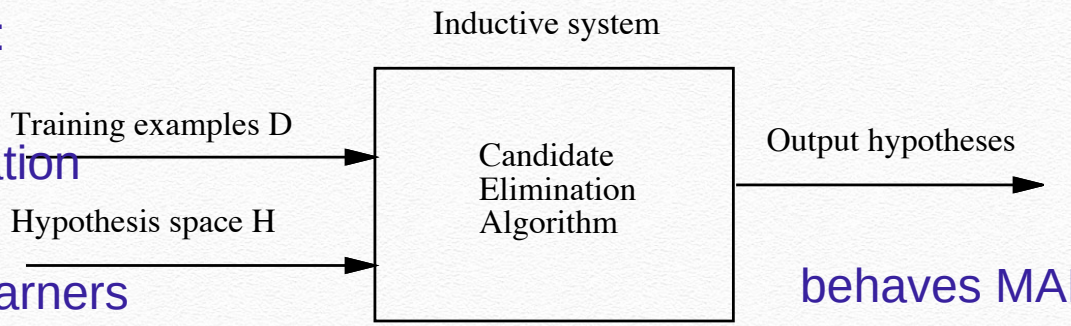


P(D) = Σ 1/ |H|

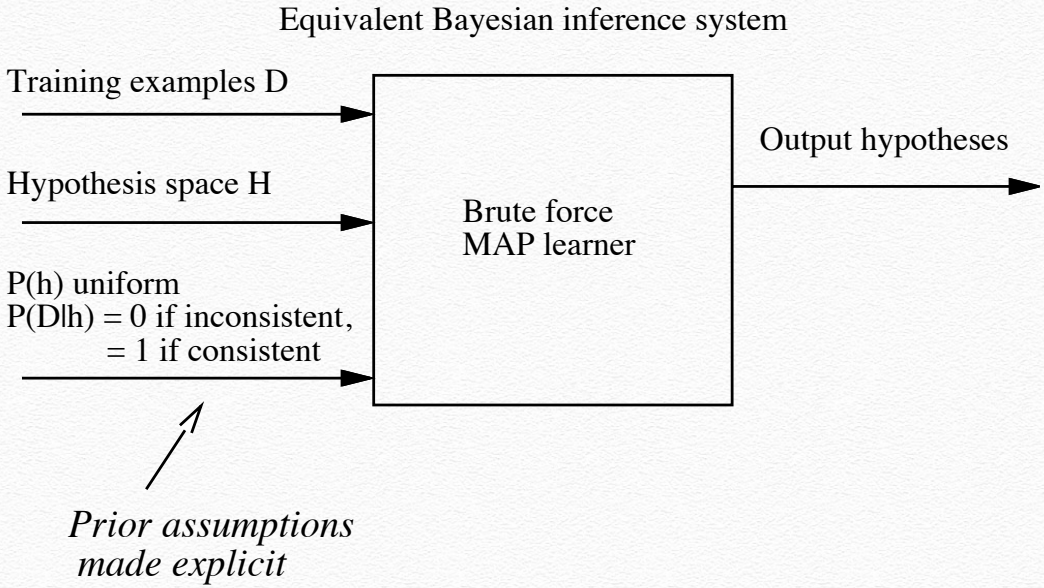
Characterizing Learning Algorithms by Equivalent MAP Learners

Concept Learning:
Find-S/
Candidate Elimination

- are consistent learners
- output MAP hypothesis



behaves MAP optimally



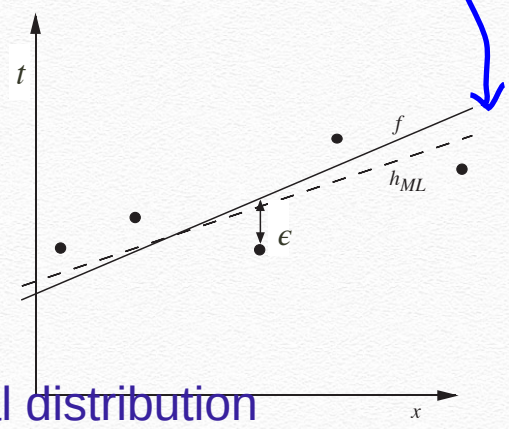
Learning a Continuous-Valued Function

f-function

latent means hidden

Consider any real-valued target function f & training examples $D = \{\langle \mathbf{x}_d, t_d \rangle\}$ where t_d is a noisy target output for training example d:

latent function



- $t_d = f(\mathbf{x}_d) + \epsilon_d$
- ϵ_d is a random noise variable drawn independently for each $\mathbf{x}_d$ according to $\epsilon_d \sim \mathcal{N}(0, \sigma^2)$

gaussian noise / normal distribution

Then, the maximum likelihood hypothesis h_{ML} is the one that minimizes sum of squared errors:

$$h_{\mathrm{ML}} = \arg\min_{h \in H} \frac{1}{2} \sum_{d \in D} \left(t_d - h(\mathbf{x}_d)\right)^2 .$$

Learning a Continuous-Valued Function

$$
\begin{aligned}
h_{\mathrm{ML}} &= \arg\max_{h \in H} p(D|h) \\
&= \arg\max_{h \in H} \prod_{d \in D} p(t_d|h) \\
&= \arg\max_{h \in H} \prod_{d \in D} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(t_d - h(\mathbf{x}_d))^2}{2\sigma^2}\right) \\
&= \arg\max_{h \in H} \sum_{d \in D} \ln \frac{1}{\sqrt{2\pi}\sigma} - \frac{(t_d - h(\mathbf{x}_d))^2}{2\sigma^2} \\
&= \arg\max_{h \in H} \sum_{d \in D} -\frac{(t_d - h(\mathbf{x}_d))^2}{2\sigma^2} \\
&= \arg\max_{h \in H} \frac{1}{2} \sum_{d \in D} -(t_d - h(\mathbf{x}_d))^2 \\
&= \arg\min_{h \in H} \frac{1}{2} \sum_{d \in D} (t_d - h(\mathbf{x}_d))^2
\end{aligned}
$$

individual likelihood for each example

assume training example is conditionally independent, given hypothesis h

N (td | h(x), sigma2)

f(xd)

P(td | f(xd))

unknown

inverted bowl